BECL606 VLSI DESIGN & Testing LAB

Part A:

1. Design a 4 Bit adder, write a verilog Description
   - verify functionality using Testbench
   - Synthesize the Design & generate the gate level Netlist.

2. Design 32 bit ALU
   - write verilog Description
   - verify functionality using Testbench
   - Synthesize the Design & generate the gate level Netlist

3. Design D-Flipflop
   - write verilog Description
   - Verify functionality using Testbench
   - Synthesize the Design & generate the gate level Netlist

4. Design JK-FF
   - write verilog Description
   - verify functionality using Testbench
   - synthesize the Design & generate the gate level Net-list

5. Design Four bit synchronous Mod-N counter
   - write verilog Description
   - verify functionality using Testbench
   - synthesize the Design & generate the gate level Netlist

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Part B

6. Construct the schematic of CMOS Inverter with load capacitance of 0.1 PF & set the width of Inverter with $W_p = 2W_n$ & generate the Layout & simulate

7. Construct the schematic of 2 input CMOS NOR gate verify the functionality & Generate the Layout.

8. Construct the schematic of the Boolean expression $Y = \overline{AB + CD + E}$ using CMOS Logic verify the functionality & Generate the Layout.

9. Construct the schematic of Common Source Amplifier with PMOS current mirror Load find the Transient response & AC response & Generate the Layout

10. Construct the schematic of Two-stage operational Amplifier verify functionality & generate the Layout.

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